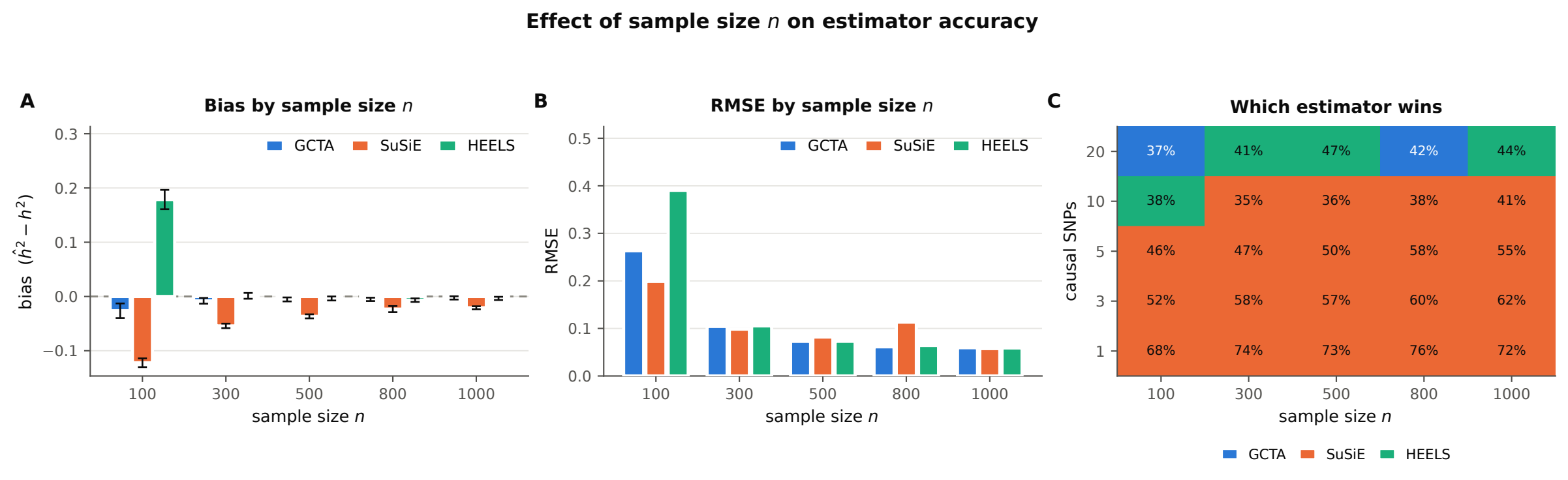
Effect of sample size n on estimator accuracy
A
Bias by sample size n
GCTA
SuSiE
HEELS
bias $(\hat{h}^2 - h^2)$
0.3
0.2
0.1
0.0
−0.1
100
300
500
800
1000
sample size n
B
RMSE by sample size n
GCTA
SuSiE
HEELS
RMSE
0.5
0.4
0.3
0.2
0.1
0.0
100
300
500
800
1000
sample size n
C
Which estimator wins
causal SNPs
20
10
5
3
1
37%
41%
47%
42%
44%
38%
35%
36%
38%
41%
46%
47%
50%
58%
55%
52%
58%
57%
60%
62%
68%
74%
73%
76%
72%
100
300
500
800
1000
sample size n
GCTA
SuSiE
HEELS